



Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: B.Tech-V & IT

Course Code & Title: CCS334 & Big Data Analytics

Name of the Faculty member (s): M.Rethinakumari

Innovative Practice Description

- **Unit / Topic: Unit III / HDFS Architecture**

- **Course Outcome:** HDFS Architecture is an important topic in Hadoop which explains the mapping and reducing concepts.

- **Activity Chosen: Flipped Class Room**

- **Justification:**

Inverting the classroom is a teaching strategy known as "flipping the classroom." This method "flipped" typical in-class instruction to better serve the needs of each individual student. HDFS Architecture explains how the data is stored and processed in Hadoop using parallel processing. In this topic students are able to understand the data types used in Hadoop and various way of storing it in the tool.

- **Time Allotted for the Activity: 20 Minutes**

- **Details of the Implementation:**

A flipped classroom is a kind of cooperative learning in which students watch videos and refer notes uploaded for their reference, take notes based on their comprehension level, and then solve the given problem using the knowledge they have learned. As a result, students' self-learning habits get better. The happening of event was intimated to the students in the class room exactly three days in prior. The supporting notes and video (<https://www.youtube.com/watch?v=PIryj2ODZN0>) were uploaded one day before. On the day of event, totally 43 students were present. Also, teams of size 4/5 were formed to discuss the concept and discussed in the class room. After discussion with the team members, one per team was called on stage to present their ideas related to Map reduce. Figure 1,2,3 & 4 shows a sample snap taken during the event and it covers three teams. Figure 5,6 shows the notes prepared by the students.

•CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO10	PO11	PSO1	PSO2	PSO3
CO3	3	2	2	2	2	3	2	3

(1 – Low 2 – Moderate 3 – High)

•PO / PSO mapped:

Innovative practice	PO1	PO2	PO5	PO10	PO11	PSO1	PSO2	PSO3
	3	2	2	2	2	3	2	3
Justification for correlation	A basic knowledge of engineering fundamentals is necessary to handle big data.	Big data storage format is important in analysis of complex engineering problems especially many business domains.	Choose appropriate big data storage databases and conduct analysis on complex problems based on the big data.	Able to communicate the decision to various analysts by understanding underlying databases.	Able to analyze a huge amount of data and make business decisions and communicate with team members.	Able to analyze huge amount of data to solve complex computation task	Able to design and analyze the data to develop reliable IT solutions	Able to determine and apply appropriate techniques for providing solutions to real world problems in Industry and Research establishments.

•Images / Screenshot of the practice:



Fig 1 & 2: Students formed group and discussed their view about the topic

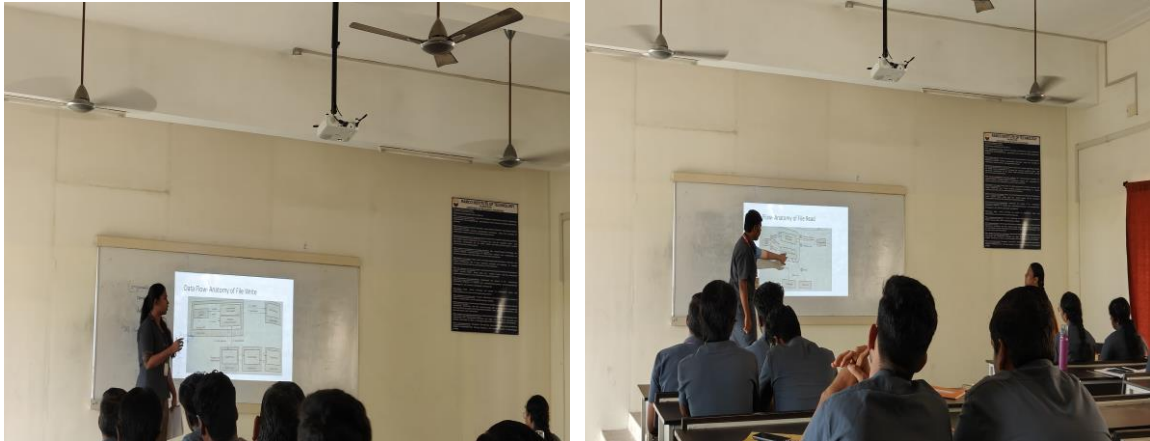


Fig 3 & 4: Students from different groups called out and explained their topic

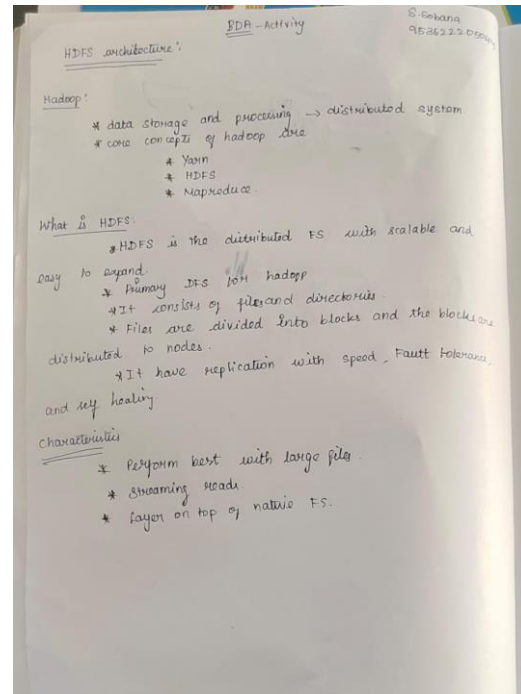
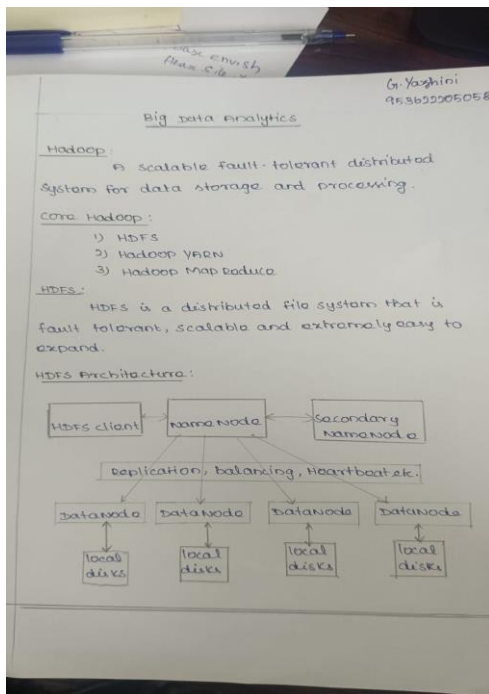


Fig 5&6: Notes prepared by the students

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- They enjoyed the freedom to learn at their own pace
- They thought that independent learning would enable them to experiment with new concepts.

❖ *Benefit of the practice:*

- Students learn at their own pace;
- There is more time for one-on-one interaction between the teacher and the student;
- There is more opportunity for student participation.
- It encourages students to arrive prepared to class.

❖ *Challenges faced in implementation:*

- If the students do not prepare for the event, then the time is entirely waster.
- The depth of the subject can be dictated by the student themselves or the group the student is working.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20i ndependently%20at%20home.>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>
- ❖ [https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GS_CS8592_Flipped_classroom.p
df](https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GS_CS8592_Flipped_classroom.pdf)
- ❖ [https://www.ritrjpm.ac.in/images/compute
r-science/2022-
2023/5.Flipped_Classroom_FDS.pdf](https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5.Flipped_Classroom_FDS.pdf)

Signature of Faculty Member

HOD